

Ender Dragon doesn't take damage from melee attacks unless hit a certain way

MC-158205 Combat

Status	Resolved / Fixed
Affected	1.14.4, 1.15.2, 20w06a, 20w09a, 20w11a, 20w22a (+52 more)
Fixed in	24w44a
Reported	2019-08-01

STEPS TO REPRODUCE

1. Launch Minecraft and create a new Creative world with default settings.
2. Open the chat and run `/summon minecraft:ender_dragon ~ ~ ~``.
3. Approach the ender dragon and face the middle of its body.
4. Perform melee attacks against the torso from the center toward the tail.
5. Perform melee attacks against the torso near the head and from different angles.
6. Perform melee attacks against the dragon's wings.
7. Fire arrows at the same body areas and compare whether the attacks register.
8. Observe that melee attacks against the torso toward the tail and against the wings sometimes do not register, while melee attacks near the head register only from certain angles and arrows register consistently.

OBSERVED BEHAVIOUR

Actual behavior

Melee attacks against the Ender Dragon do not consistently register as hits.

On the dragon's central body hitbox, attacks aimed at the middle or rear of the model generally fail to connect. Hits may register only when aimed near the head end of this hitbox and from certain angles. The exact result appears to depend on the relative position and orientation of the dragon and the player. The head and tail hitboxes appear to register melee attacks normally.

The dragon's wing hitboxes do not register melee attacks at all. This affects attacks made while facing the wings as well as attacks aimed toward the middle or rear of the dragon.

Projectile attacks, such as arrows, consistently register against the affected parts of the dragon. This indicates that the issue is specific to melee hit detection rather than the dragon being immune to damage.

EXPECTED BEHAVIOUR

Melee attacks against the ender dragon should consistently register and deal damage whenever the player strikes a valid part of its body. Hits to the head, torso, tail, and wings should work regardless of the dragon's orientation, the player's position, or the attack angle, rather than only registering near the head. Damage detection should behave reliably across the entire dragon model.

ENVIRONMENT

Minecraft Java Edition Versions: 1.14.4, 1.15.2, 20w06a, 20w09a, 20w11a, 20w22a, 1.16 Pre-release 8, 1.16.2, 1.16.3, 1.16.4, 20w45a, 20w46a, 20w51a, 1.16.5, 21w08a, 21w18a, 1.17 Pre-release 3, 1.17, 1.17.1 Pre-release 1, 1.17.1, 21w38a, 21w42a, 1.18 Pre-release 1, 1.18, 1.18.1 Pre-release 1, 1.18.1, 1.18.2, 22w12a, 22w16b, 1.19 Pre-release 2, 1.19 Pre-release 4, 1.19, 1.19.2, 1.19.3 Pre-release 2, 1.19.3 Pre-release 3, 1.19.3, 1.19.4 Pre-release 3, 1.19.4, 23w18a, 1.20 Release Candidate 1, 1.20, 1.20.1, 23w31a, 23w32a, 1.20.2 Release Candidate 2, 1.20.2, 23w40a, 23w43a, 1.20.4, 23w51b, 24w05b, 24w06a, 24w07a, 24w14a, 1.20.5 Pre-Release 1, 1.20.5, 1.20.6 Release Candidate 1, 1.21 (includes snapshots)

ORIGINAL DESCRIPTION

When fighting the ender dragon, melee attacks sometimes don't register at all.

On the middle hitbox where the torso is ublicated, hits don't register from the center of the model towards the tail. Instead, they are only registering near the head, and only on certain weird angles. From what i've seen, the other

hitboxes where the head and tail are located seem to be unaffected. Another important thing to say is that wings don't ever register hits (more info in the comments).

Basically, melee attacks done facing to the back or middle of the model don't register, seems to depend on the dragon's or player's position. Arrows however, always register.

Code Analysis

Code analysis by insane96mcp can be found in this comment.

This issue appears to be caused by two root problems:

- There is a desync between client and server for entity IDs for the small hitboxes the dragon is composed of due to an off-by-one error (see

MC-274526

Resolved

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- The hit detection used by the game does not account for the fact that due to its size, the dragon entity can be located in a chunk the player is not looking at (see

MC-261638

Open

, also affects other entities with large hitboxes)

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-158205>